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November 09, 2023

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

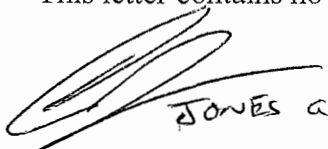
SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387(388)/2023-003-00
UNIT 1 LICENSE NO. NPF-14
UNIT 2 LICENSE NO. NPF-22
PLA-8093

Docket No. 50-387
50-388

Attached is Licensee Event Report (LER) 50-387(388)/2023-003-00. The LER reports a condition in which the D Diesel Generator was slow to start and reach required frequency within acceptance criteria. This condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by Technical Specifications.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

 *JONES acting sup for E. Casulli*

E. Casulli

Attachment: LER 50-387(388)/2023-003-00

Copy: NRC Region I
Mr. C. Highley, NRC Senior Resident Inspector
Ms. A. Klett, NRC Project Manager
Mr. M. Shields, PA DEP/BRP



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Susquehanna Steam Electric Station (SSES) Unit 1

☒ 050
☐ 052

2. Docket Number

00387

3. Page

1 OF 3

4. Title

D Diesel Generator Inoperable due to Clogged Fuel Injectors

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☒ 050	Docket Number
09	12	2023	2023	- 003 -	00	11	09	2023	SSES Unit 2	☒ 050	00388
									Facility Name	☐ 052	Docket Number

9. Operating Mode	1	10. Power Level	100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Derek Smith – Nuclear Regulatory Affairs

Phone Number (Include Area Code)

570-542-1377

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
D	EK	DG	C634	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month Day Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On September 12, 2023, at approximately 05:01, during the performance of surveillance testing, the D Diesel Generator (DG) failed to reach the required frequency within the acceptance criteria specified in Surveillance Requirement (SR) 3.8.1.7. Operators declared SR 3.8.1.7 failed and entered Technical Specification (TS) Limiting Condition of Operation (LCO) 3.8.1, Condition B, on both Susquehanna Steam Electric Station Unit 1 and 2 for the inoperable D DG. The other three DGs (A, B, and C) were verified to be OPERABLE and operators substituted the E DG in for the D DG. After completion of the E DG substitution, LCO 3.8.1, Condition B, was exited on September 12, 2023, at approximately 22:52.

The failure to reach the required frequency within the acceptance criteria was the result of clogged fuel injectors. The condition of the inoperable D DG was determined to likely have existed for longer than allowed by TS, and the event was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS.

There were no actual consequences to the health and safety of the public as a result of this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Susquehanna Steam Electric Station (SSES) Unit 1	☒ 050	2. DOCKET NUMBER 05000387	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 003 -	REV NO. 00

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power (RTP)

Unit 2 – Mode 1, approximately 93 percent RTP

There were no structures, systems, or components that were inoperable at the start of the event and contributed to the event.

EVENT DESCRIPTION

On September 12, 2023, at approximately 05:01, operators started the D Diesel Generator (DG) [EIS Code/Component Code: EK/DG] in accordance with the applicable surveillance procedure. The D DG reached the required frequency at a time of 12.6 seconds, which exceeds the acceptance criteria of 10 seconds specified in Surveillance Requirement (SR) 3.8.1.7. Operators declared SR 3.8.1.7 failed and entered Technical Specification (TS) Limiting Condition of Operation (LCO) 3.8.1, Condition B, on both Susquehanna Steam Electric Station (SSES) Unit 1 and 2 for the inoperable D DG. The other three DGs (A, B, and C) were verified to be OPERABLE and operators substituted the E DG in for the D DG. After completion of the E DG substitution, LCO 3.8.1, Condition B, was exited on September 12, 2023, at approximately 22:52.

The condition of the inoperable D DG was determined to likely have existed for longer than allowed by TS, and the event was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS.

The following is a timeline of events:

February 2023 to August 2023 – The D DG has higher than normal accumulated low load run time, which was considered as a possible contributor to the clogged injectors. However, start times were acceptable after the low load operation, and there was not an increasing trend in start time.

08/17/2023 – D DG monthly surveillance start time was 8.65 seconds.

09/12/2023 – D DG monthly surveillance start time was 12.6 seconds. LCO 3.8.1, Condition B, was entered at approximately 05:01 due to the D DG being inoperable. The E DG was substituted for the D DG and LCO 3.8.1, Condition B, was exited at approximately 22:52.

09/21/2023 – Troubleshooting identified fuel injectors had excessive carbon buildup. The fuel injectors were cleaned.

09/22/2023 – D DG monthly surveillance was performed successfully with a start time of 7.95 seconds. D DG was substituted back in for the E DG and the D DG was declared operable at approximately 02:10 on 09/23/2023.

SSES TS LCO 3.8.1 requires, in part, that four DGs are OPERABLE in MODES 1, 2, and 3. Condition B requires that an inoperable DG be restored to OPERABLE status within 72 hours or that each unit be shut down to MODE 3 in 12 hours and MODE 4 in 36 hours. Based on the identified cause of the D DG slow start, this condition likely existed for longer than allowed by TS. As a result, this condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME

Susquehanna Steam Electric Station (SSES)
Unit 1

**050****052****2. DOCKET NUMBER**

05000387

3. LER NUMBER**YEAR**

2023

SEQUENTIAL

NUMBER
- 003 -

REV

NO.
00

NARRATIVE**CAUSE OF EVENT**

The cause of the slow start was clogged fuel injectors.

ANALYSIS/SAFETY SIGNIFICANCE

As described in Section 8.3 of the Susquehanna Updated Final Safety Analysis Report (UFSAR), and required by 10 CFR 50, Appendix A, General Design Criterion 17, a single failure of a DG will not prevent safe shutdown of the units in the event of a design basis accident. While the D DG was inoperable, the three remaining DGs (A, B, and C) were aligned and operable. Therefore, the slow start of the D DG did not prevent the DGs from fulfilling their safety function.

The safety function of the DGs is to provide power to equipment as assumed in the safety analyses described in Chapter 15 of the Susquehanna UFSAR. In this case, the safety analyses assume the DGs will start in 25.1 seconds of a valid start signal. Additionally, the loading analysis for the DGs assume the Residual Heat Removal motors are loaded at 13 seconds to ensure the DG can handle the load of the large motors. The start time of 12.6 seconds observed during the September 12, 2023, surveillance test was within that assumed in the accident analyses. Therefore, the D DG was able to perform its safety function.

Based on engineering analysis, this event will not be counted as a safety system functional failure under the NRC Reactor Oversight Process Performance Indicator.

CORRECTIVE ACTIONS

Key corrective actions included all D DG fuel injectors being cleaned. Although the station had implemented vendor guidance for low load run time on the DG, Susquehanna is also taking additional actions to refine guidance on low load run times for the DGs.

COMPONENT FAILURE INFORMATION

Component Name – Diesel Generator 'D'

Component Identification – 0G501D

Manufacturer – Cooper Energy Services

PREVIOUS OCCURRENCES

None.